
THEMATIC RESEARCH · BITCOIN × ARTIFICIAL INTELLIGENCE · 2026–2030

**The AI economy's Bitcoin TAM is option value:
\$1.5–5 trillion of 2030 forecasts against roughly \$50
million of settled reality.**

Version 1.1 · All claims independently verified against primary sources, 10 June 2026 ·

\$1.5–5T

FORECAST AGENTIC-
COMMERCE VALUE, 2030
(JUNIPER; MCKINSEY)

~\$50M

CUMULATIVE CRYPTO-
SETTLED AGENTIC VOLUME
THROUGH APR 2026

>\$70B

CONTRACTED AI/HPC
REVENUE ACROSS PUBLIC
BITCOIN MINERS

Executive summary

The "AI pays in Bitcoin" thesis is real but pre-revenue. Forecasts put agentic commerce at \$1.5–5 trillion of annual value by 2030^{[1][2]}; the leading live payment rail for AI agents has settled roughly \$50 million — cumulatively — through April 2026^{[7][17]}. That rail is not Bitcoin: Coinbase's x402 protocol settles in stablecoins and was institutionalized under the Linux Foundation in April 2026 with Stripe, Cloudflare, Shopify, and Solana as founding members^{[5][6]}. Bitcoin's native agent stack — L402, Lightning agent tools, identity-free pay-per-call — shipped in February 2026 and is technically complete, with no published volume^{[3][4]}.

Two structural developments define the current state of play. First, the convergence path opened: Tether's USDT went live on Lightning via Taproot Assets in March 2026, explicitly marketed for AI-agent payments — meaning "stablecoins win" and "Bitcoin rails win" are no longer mutually exclusive outcomes^{[13][14]}. Second, the one channel where AI and Bitcoin already exchange contracted dollars is infrastructure, not payments: more than \$70 billion of AI and high-performance-computing contracts have been announced across the public Bitcoin mining sector, and the resulting capacity migration has pushed network hashrate into the longest sustained drawdown of the industrial mining era^{[10][11]}. The AI-treasury channel, by contrast, is confirmed empty: no AI company holds Bitcoin per filings, and the measurable corporate flow runs the other way — miners selling Bitcoin to fund AI conversions^{[11][15]}.

The Fund tracks five signposts on this theme (Exhibit 5). One is partially triggered. This note presents the analysis; it does not constitute investment advice or a recommendation regarding

any security or digital asset.

Three transmission channels

C1 — Payments. AI agents settle machine-to-machine commerce on Bitcoin or Lightning, creating fee demand, channel liquidity, and monetary-premium effects.

C2 — Treasury. AI companies or sovereigns accumulate Bitcoin as the machine economy's reserve asset.

C3 — Infrastructure. Bitcoin miners convert power, land, and cooling into AI compute — value accrues to converting equities and is neutral-to-negative for Bitcoin's own network fundamentals.

As of June 2026: C1 is pre-revenue, C2 is empty, and C3 carries all of the contracted dollars.

The forecasts — and why they overstate the settlement opportunity

Data: Juniper Research forecasts \$1.5 trillion of agentic commerce spend in 2030, from a 2025–26 base it characterizes as pilot deployments only (secondary coverage attributes a ~\$8 billion 2026 baseline — an implied ~187× ramp in four years)^[1]. McKinsey's October 2025 study sizes agentic commerce at \$3–5 trillion globally by 2030, including up to \$1 trillion of orchestrated US retail revenue^[2].

Assessment: McKinsey's operative verb is "orchestrate," not "settle." Commerce an agent arranges will mostly clear on cards and ACH behind the agent interface. The settlement-layer opportunity is the relevant pool for crypto rails, and at crypto take-rates of 0.1–1.0% (versus 200–300bps for cards), even \$3 trillion of agentic volume implies a \$3–30 billion annual settlement-revenue pool — material, but two orders of magnitude below the headline figures commonly quoted.

EXHIBIT 1 · THIRD-PARTY ESTIMATES AND METHODOLOGY ASSESSMENT

SOURCE	FIGURE	METHOD	ASSESSMENT
Juniper Research (Apr 2026) ^[1]	\$1.5T in 2030	Top-down adoption ramp from ~\$8B 2026 base	~187× implied four-year ramp; no rail-share split; volume ≠ settlement revenue
McKinsey (Oct 2025) ^[2]	\$3–5T by 2030	Top-down share of global commerce	"Orchestrated" ≠ settled — most volume clears on legacy rails behind the agent UX
Chainalysis (Jun 2026) ^[7]	>100M txns	Bottom-up on-chain count	~Half artificial per Artemis ^[9] ; counts ≠ value (~\$50M cumulative ^[17])

The live rails: a stablecoin incumbent, a Bitcoin challenger, and a bridge

x402 — the incumbent

Data: x402 (Coinbase-developed, HTTP-402-based, USDC settlement) went from near zero in mid-2025 to over 100 million cumulative transactions by Q1 2026; by April 2026, coverage cites 165 million-plus transactions, roughly \$50 million of cumulative volume, and 480,000-plus transacting agents^{[7][17]}. Daily run-rate as of March 2026: ~131,000 transactions generating ~\$28,000 of volume, at an average payment of ~\$0.20^[8]. Artemis Analytics attributes roughly half of observed transactions to artificial activity — self-dealing and wash patterns^[9]; early growth was inflated by memecoin minting^[7]. Payment quality is nonetheless improving: \$1-plus payments rose from 49% to 95% of volume in early 2026, and tester-to-payer conversion improved four-fold in six months^{[7][17]}. The x402 Foundation now sits under the Linux Foundation (April 2026) with settlement on Base, Ethereum, Solana, and Stellar — and no Bitcoin or Lightning support announced^[6].

L402 / Lightning — the challenger

Data: Lightning Labs open-sourced its Lightning agent toolkit in February 2026: seven composable skills covering node operation, key-isolated remote signing, scoped spending credentials, paying and hosting L402-gated APIs, and node queries over MCP; the *Inget* client pays a Lightning invoice on any HTTP 402 challenge with no account, API key, or identity, and pairs with the Aperture reverse proxy to close a full agent-to-agent commerce loop^{[3][4]}. Coinbase launched its x402-based Agentic Wallets the same day^[17]. **Assessment:** the Bitcoin-native stack is technically complete and its identity-free settlement is a genuine differentiator — agents do not have government IDs or bank accounts — but sixteen weeks post-launch there is no published L402 volume data, a gap we re-confirmed by direct search in June 2026. Public Lightning channel-capacity data for 2025–26 is internally contradictory across sources (Exhibit 2) and is flagged for primary-source reconciliation.

Taproot Assets — the bridge

Data: USDT went live on Lightning via Taproot Assets around March 2026 on a production-grade protocol (auditable issuer supply, multi-channel payments), per Tether CEO confirmation — though with no published supply or volume figures, no major exchange distribution, and a thin wallet ecosystem^[13]. Tether and Bitfinex explicitly market the rail for AI-agent payments via L402^[14]. **Assessment:** this is the development that merges the scenarios — stablecoin-denominated agent payments settling over Bitcoin infrastructure. It is live, funded, promoted, and pre-volume. Notably, the two agent-payment ecosystems remain unmerged: the Bitcoin camp's standard is L402 while x402 excludes Bitcoin entirely.

EXHIBIT 2 · KEY METRICS

METRIC	VALUE	AS OF	STATUS
Agentic commerce forecast, 2030 (Juniper) ^[1]	\$1.5T (from ~\$8B 2026 base)	Apr 2026	Verified (base single-sourced)
Agentic commerce forecast, 2030 (McKinsey) ^[2]	\$3–5T	Oct 2025	Verified
x402 cumulative transactions / volume / agents ^[17]	165M+ / ~\$50M / 480k+	Apr 2026	Verified
x402 daily throughput ^[8]	~131k txns · ~\$28k · ~\$0.20 avg	Mar 2026	Verified
x402 artificial-activity share (Artemis) ^[9]	~50% of transactions	Mar 2026	Verified
L402 / Lightning agent-payment volume ^{[3][4]}	No published data	Jun 2026	Confirmed gap
Lightning public channel capacity ^[16]	4,100–5,637 BTC reported, late '25–'26	Jun 2026	Contested
USDT on Lightning (Taproot Assets) ^[13]	Live; no volume/exchange data	Mar 2026	Verified (single-source date)
Public-miner contracted AI/HPC revenue ^[11]	>\$70B cumulative	Q1 2026	Verified
Bitcoin network hashrate, 30-day MA ^[10]	964 EH/s · – 13.2% vs peak · 187-day drawdown	18 May 2026	Verified, corroborated
AI/HPC equity value per MW ^[10]	~3× since summer 2025	18 May 2026	Verified
AI-company Bitcoin treasury holdings ^[15]	Zero confirmed	Jun 2026	Verified (absence)

The treasury channel is empty — and currently flows backward

Data: A June 2026 filings sweep found no AI company holding Bitcoin in corporate treasury. The closest adjacency is legacy: SpaceX — owner of xAI since February 2026 — holds 18,712 BTC (~\$1.29 billion) per its IPO filing, purchased in 2021 with no AI rationale stated^[15]. The US Strategic Bitcoin Reserve remains undisclosed in size, framed by officials entirely as custody and store-of-value — no official has invoked AI. The one measurable AI-related corporate Bitcoin flow

is miners selling treasury Bitcoin to fund data-center conversions^[11]. **Assessment:** we assign this channel no value in the base case and monitor for a first mover.

The infrastructure channel carries the contracted dollars

Data: More than \$70 billion of cumulative AI/HPC contracts have been announced across public Bitcoin miners, including CoreWeave–Core Scientific at \$10.2 billion over 12 years (a contractual relationship — the acquisition attempt failed in an October 2025 shareholder vote), TeraWulf at \$12.8 billion of contracted HPC revenue, and Hut 8's \$7 billion, 15-year River Bend lease, where 2026 coverage identifies Fluidstack and Anthropic as counterparties^{[11][12]}. The eleven largest US public miners shed ~7 EH/s of hashrate in a single quarter as power moved to hyperscalers on 10–15-year leases; the per-megawatt equity value of AI capacity has roughly tripled since summer 2025; and network hashrate sits 13.2% below its November 2025 peak in a 187-day drawdown — the longest of the industrial mining era, with a further ~9% difficulty decline estimated for mid-June 2026^[10].

Assessment: this is the only AI×Bitcoin channel with contracted dollars, and it cuts both ways — substantial value creation at converting operators against erosion of Bitcoin's security-budget inputs. Converting miners are progressively becoming AI-infrastructure businesses whose economics decouple from Bitcoin. The Anthropic/River Bend datapoint is the first direct AI-lab footprint on former Bitcoin-mining infrastructure; it is compute demand, not Bitcoin demand.

EXHIBIT 3 · US PUBLIC MINER HASHRATE, Q4 2025 → Q1 2026 (VANECK, COMPANY DISCLOSURES)

COMPANY	Q4 2025 (EH/S)	Q1 2026 (EH/S)	CHANGE	STATED AI/HPC POSTURE
IREN	43.0	35.8	-17%	Multi-year mining exit committed
Keel (KEEL)	19.5	12.0	-38%	Furthest along; mining capex halted
Cipher (CIFR)	15.6	11.1	-29%	Full exit by July 2027 (outside date)
TeraWulf (WULF)	6.8	4.0	-41%	Exit by ~April 2028 halving
Core Scientific (CORZ)	10.9	9.7	-11%	Near-pure-play AI by early 2027
Riot (RIOT)	34.0	42.3	+8.3 EH/s	Partial AI pivot; still growing hashrate
Bitdeer (BTDR)	43.2	50.2	+7.0 EH/s	Partial AI pivot; still growing hashrate
MARA	51.9	55.5	+3.6 EH/s	Partial AI pivot; still growing hashrate
Top-11 total	293.9	287.1	-6.8 EH/s	Structural capacity migration ^[10]

Scenario analysis

Probabilities are TON618 Capital analyst judgment as of 10 June 2026, not market-implied. Scenarios are analytical constructs for assessing the theme; they are not forecasts of, or guides to, any portfolio action.

EXHIBIT 4 · 2030 SCENARIOS, PROBABILITY-WEIGHTED

SCENARIO	PROB.	KEY ASSUMPTIONS	2030 OUTCOME FOR "AI ON BITCOIN"
Bull	15%	Agentic volume $\geq$ \$3T; identity-free settlement proves decisive; Taproot-Assets stablecoins scale on Lightning	Bitcoin rails carry 5–10% of agentic settlement (\$150–300B); \$1.5–3B settlement-revenue pool. Caveat: if the bull case arrives via stablecoins-on-Bitcoin-rails, value accrues to Bitcoin <i>infrastructure</i> while the unit of account stays USD
Base	55%	Agentic volume $<$ \$1T by 2030; stablecoins take $\geq$ 90% of crypto-settled agent payments on x402-ecosystem chains; miner conversion continues	Bitcoin payments leg stays niche ($<$ \$50B); AI $\times$ Bitcoin value accrues through the infrastructure channel
Bear	30%	Agentic payment demand stays negligible (extrapolating current run-rates); rail ambiguity resolves negatively; hashrate drawdown extends	Payments $\approx$ 0; treasury stays at zero; infrastructure conversion proceeds but fully decouples miners from Bitcoin; security-budget narrative erodes without an offsetting demand channel

Weights unchanged from v1.0: this cycle's verification moved evidence roughly symmetrically (pro-Bitcoin-rails: USDT live on Lightning with explicit agent positioning; pro-stablecoin-incumbency: Linux Foundation institutionalization, zero treasury demand, no L402 volume).

Risk factors

EXHIBIT 5 · PRINCIPAL RISKS TO THE THEME

RISK	LIKELIHOOD	MAGNITUDE	NOTE
Stablecoin rails win agent payments outright	High	High	Linux Foundation governance; Stripe/Shopify distribution ^[6] . Partial offset: Taproot-Assets bridge ^[13]
Agentic-commerce volumes disappoint vs. forecasts	Med-High	High	~\$50M cumulative settled vs. \$1.5–5T forecasts ^{[8][17]}
Security-budget narrative damage from miner exodus	Medium	Medium	Record drawdown corroborated; further difficulty decline due June 2026 ^[10]
Regulatory scrutiny of identity-free agent payments	Medium	Med-High	L402's differentiator is also its AML surface
Adoption metrics inflated by artificial activity	High	Medium	~50% of x402 transactions artificial ^[9] ; mitigant: payment quality improving ^[7]
Execution and counterparty concentration at converting miners	Medium	High	Fluidstack appears across multiple miner contracts ^{[11][12]} ; failed CoreWeave acquisition shows deal risk ^[12]

What we are watching

EXHIBIT 6 · SIGNPOSTS, AS OF 10 JUNE 2026

#	SIGNPOST	STATUS
S1	Published L402/Lightning agent-payment volume exceeding \$1M/day	Not observed; data gap re-confirmed Jun 2026
S2	A major AI lab integrates Lightning/Bitcoin settlement in production	Not observed (River Bend involvement is compute, not payments)
S3	x402 adds Bitcoin/Lightning settlement, or Taproot-Assets stablecoins reach scale on Lightning	Partially triggered — USDT live Mar 2026, pre-volume; x402 still excludes Bitcoin
S4	Lightning public capacity inflects upward for two consecutive quarters	Indeterminate — capacity data contested; primary reconciliation in progress
S5	Wash-adjusted crypto-settled agentic volume crosses \$10B annualized	Not observed; ~\$50M cumulative through Apr 2026

Sources

1. Juniper Research, "Agentic Commerce Set to Generate \$1.5 Trillion Globally by 2030," press release, 7 Apr 2026.
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3. Lightning Labs, "The Agents Are Here and They Want to Transact," 11 Feb 2026.
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6. x402 Foundation establishment under the Linux Foundation, 2 Apr 2026 (founding members incl. Stripe, Cloudflare, Shopify, Solana); x402.org ecosystem documentation.
7. Chainalysis, "x402 and agentic payments adoption," 3 Jun 2026.
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9. Artemis Analytics, x402 wash-activity analysis (30-day filter), 11 Mar 2026.
10. VanEck, "Bitcoin ChainCheck," data as of 18 May 2026; corroboration: CoinDesk/CoinShares, "Bitcoin Hashrate Posts First Quarterly Drop in Six Years," 30 Mar 2026.
11. CoinDesk, "Bitcoin Miners Are Becoming AI Companies — and Selling Their BTC to Fund the Transition" (CoinShares data), 27 Mar 2026.
12. Core Scientific Form 8-K, Q1 FY2026 (SEC EDGAR); Sherwood News on the CoreWeave shareholder vote, Oct 2025; TeraWulf Form 8-K, Q1 2026 (SEC EDGAR).

13. Tether, USDT on Lightning announcement, 30 Jan 2025; go-live coverage citing CEO confirmation, Mar 2026; Lightning Labs, Taproot Assets v0.7, 16 Dec 2025.
 14. Bitfinex, "Why Bitcoin and Stablecoins on Lightning Will Power the Next Phase of AI Agent Payments," 6 Mar 2026.
 15. CoinDesk, "SpaceX Holds 18,712 Bitcoin at Fair Value of \$1.29B, IPO Filing Shows," 20 May 2026; SEC EDGAR filings sweep (Nvidia FY2026 et al.), Jun 2026.
 16. CryptoSlate (Aug 2025), news.bitcoin.com, and BYDFi (May 2026) — Lightning capacity series (figures conflict; flagged contested).
 17. Coinbase, Agentic Wallets launch, 11 Feb 2026; ecosystem adoption trackers, Apr 2026.
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